



Deep learning methods in digital forensics

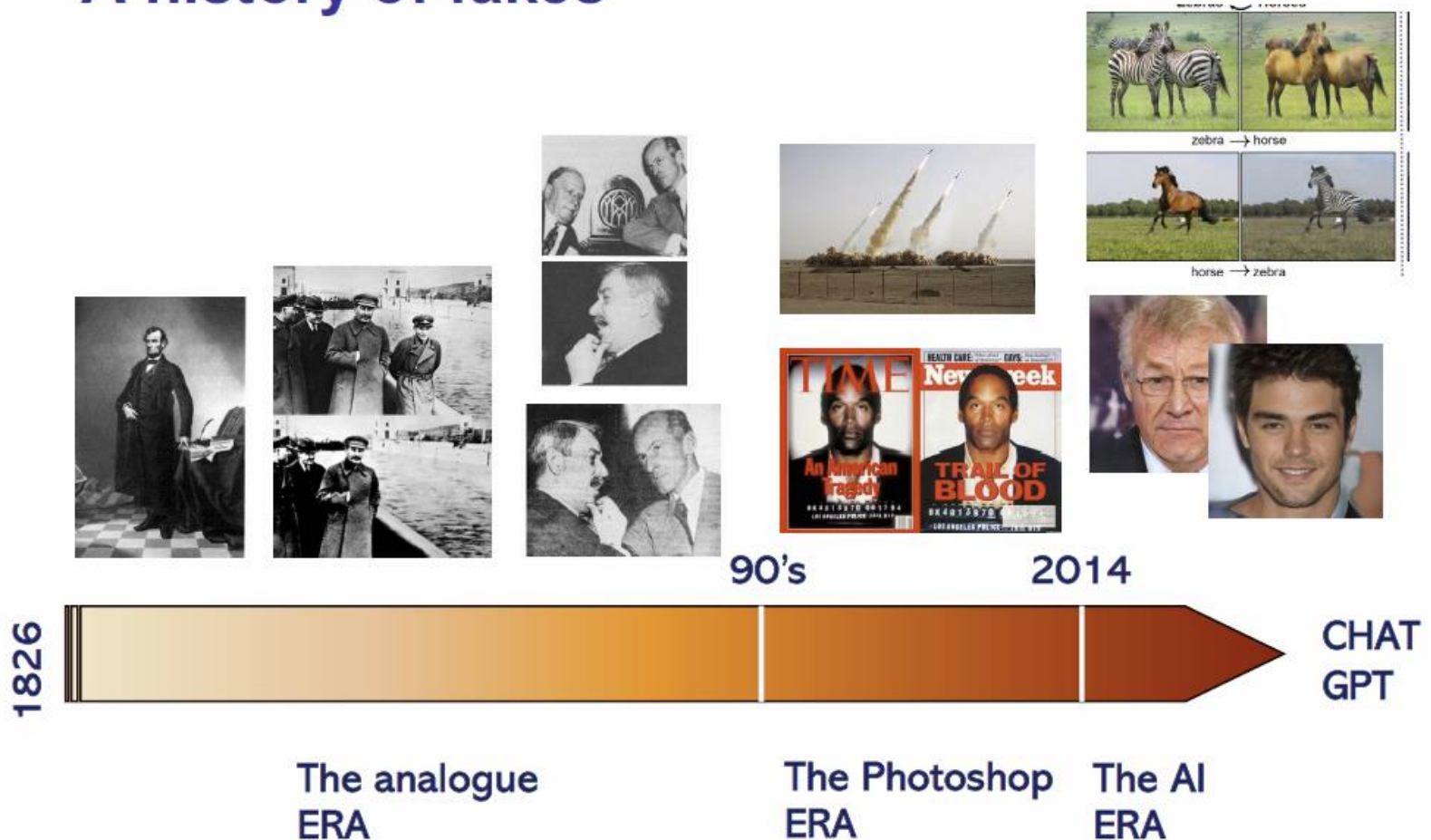
Taras Holotyak

Agenda

- Introduction
- AI media-revolution
- DNN watermarking in a nutshell
- GAN watermarking: early and new solutions
- Conclusions

Introduction

A history of fakes



Introduction

When problem happen

In the end of the 90's, the diffusion of easy-to-use image editing tools raised increasing alarm about the credibility of digital images and the possible use of manipulated images for malevolent purposes

- **Use of digital images in a court of law**
- **Gossip / defamation**
- **Bias the political debate**

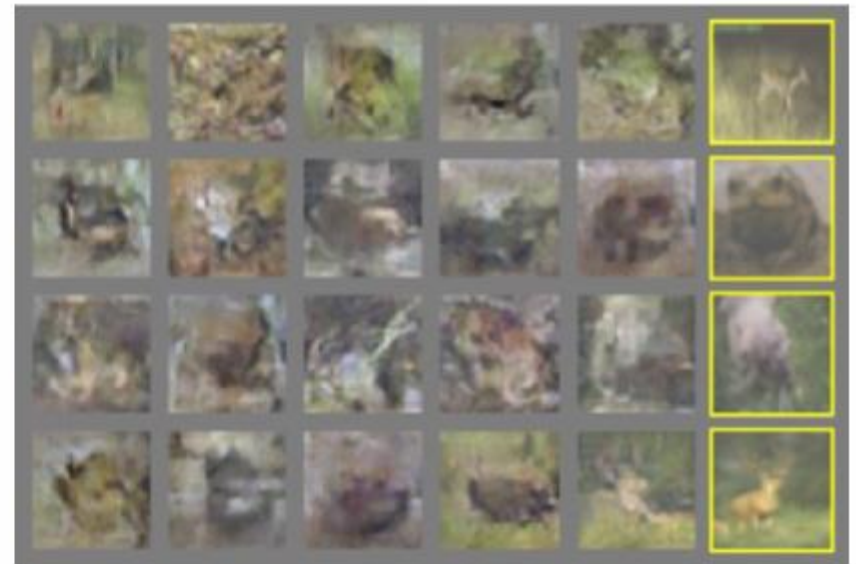
First technical solutions were appeared in the early 2000's proposed by researchers working in steganalysis

AI media-revolution

AI revolution

The AI revolution of photo-editing started in 2014

*Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., ... & Bengio, Y. (2014). Generative adversarial nets. Advances in neural information processing systems, 27 (this paper is receiving more than 1000 citations **per month**)*



AI media-revolution

Further evolution of AI revolution



AI media-revolution

Further evolution of AI revolution



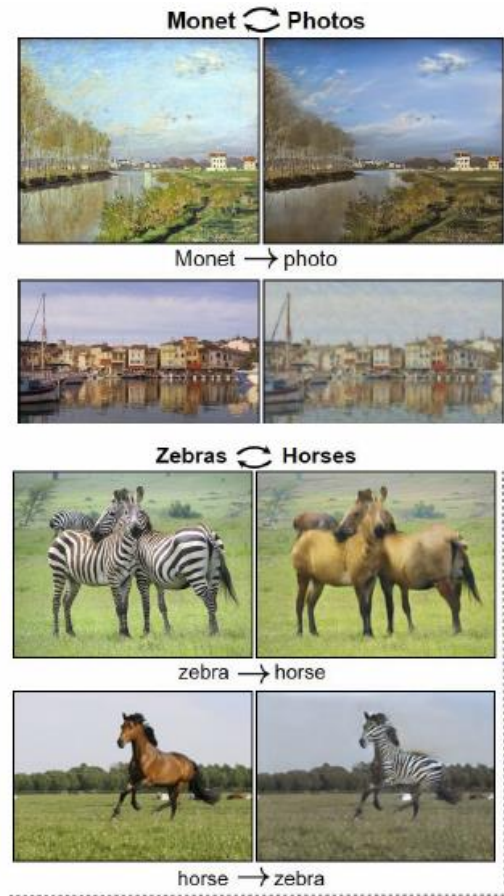
AI media-revolution

This objects do not exist



AI media-revolution

Im2Im translations



AI media-revolution

Examples of AI video

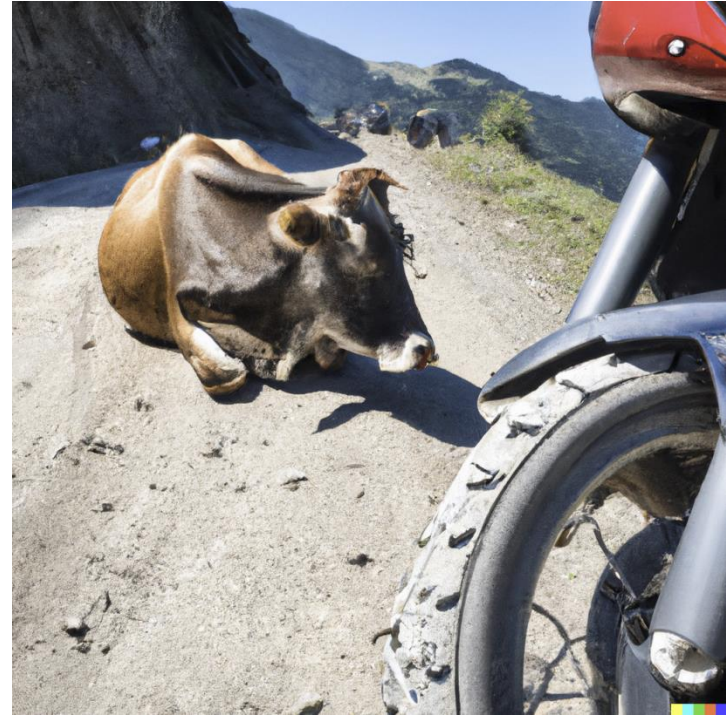


AI media-revolution

Chat-GPT, DALL-E



Two Italian girls eating pizza
un a sunny day



A motorbike blocked by a cow on
a mountain road

AI media-revolution

An ubiquitous presence

- Image taken by cell-phone are “doctored” at the origin
- All cell phones have AI-based apps to retouch/manipulate images easily and in a credible way
- AI image editing tools are freely available online
- All photo editors are equipped with AI tools easier and easier to use
- Video enacting or puppeteering will soon be available on any video conferencing systems
- Companies are already working on a video version of DALL-E

AI media-revolution

Does conventional MMF still make sense?

- The amount of possible modifications, their extent and the progress of image editing AI clearly outgun the capabilities of MMF research
- MMF forensics **IN THE WILD** is more difficult than ever
- Understanding that an image has been “manipulated” by an AI tool may (no longer) be a meaningful info

DNN watermarking

Shift of paradigm



An old
idea

Why don't we embed an invisible watermark in all AI images so to ease

- Origin verification
- Trace manipulation history
- Detection of abuses

Unfeasible solution: how can we enforce the watermarking of all images generated or edited by means of AI?

DNN watermarking

Shift of paradigm



A new
idea

Rather than engaging a hopeless race of arms with generative AI companies, team up with them to ease the ethical use of digital images and identify abuses

Watermark AI generative models so that all the images they produce contain a watermark

Possibly feasible: only a handful of companies are capable to train *from scratch* a new AI generative model

DNN watermarking

DNN watermarking in nutshell

DNN watermarking has been proposed as a way to protect the IPR of DNN models

Y. Uchida, Y. Nagai, S. Sakazawa, and S. Satoh, "Embedding watermarks into deep neural networks," in Proc. ICMR'17, 2017

Li, Y., Wang, H., & Barni, M. (2021). A survey of deep neural network watermarking techniques. Neurocomputing, 461, 171-193.

It indissolubly embeds within a DNN model a piece of information to be used later for a given purpose

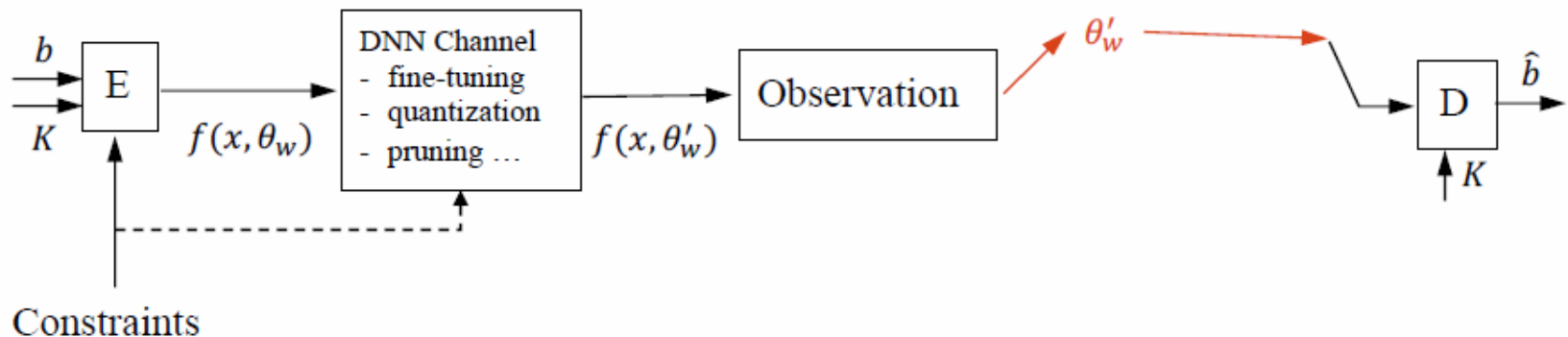
The watermark should resist moderate to strong model modifications, like pruning, compression, fine tuning and even transfer learning



DNN watermarking

DNN watermarking = function watermarking

Digital communication problem (multi-bit watermarking)

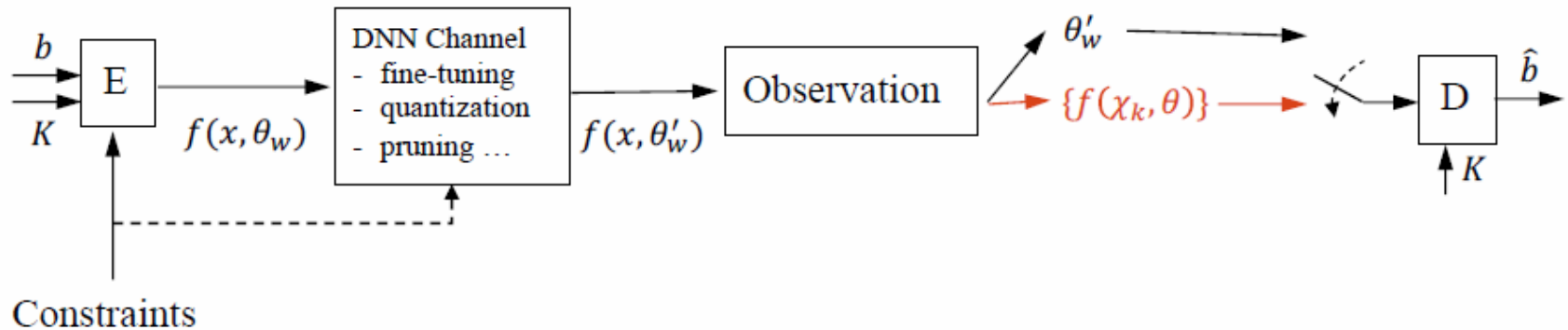


White-box watermarking

DNN watermarking

DNN watermarking = function watermarking

Digital communication problem (multi-bit watermarking)

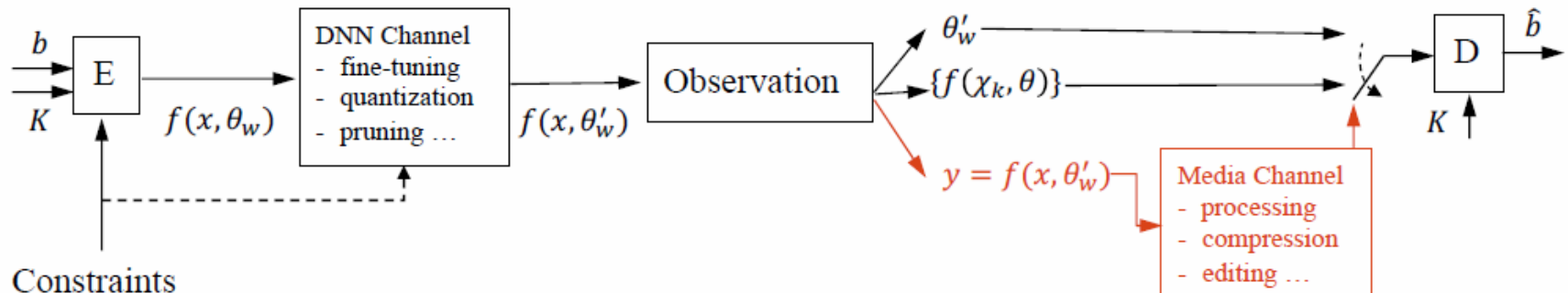


Black-box watermarking

DNN watermarking

DNN watermarking = function watermarking

Digital communication problem (multi-bit watermarking)



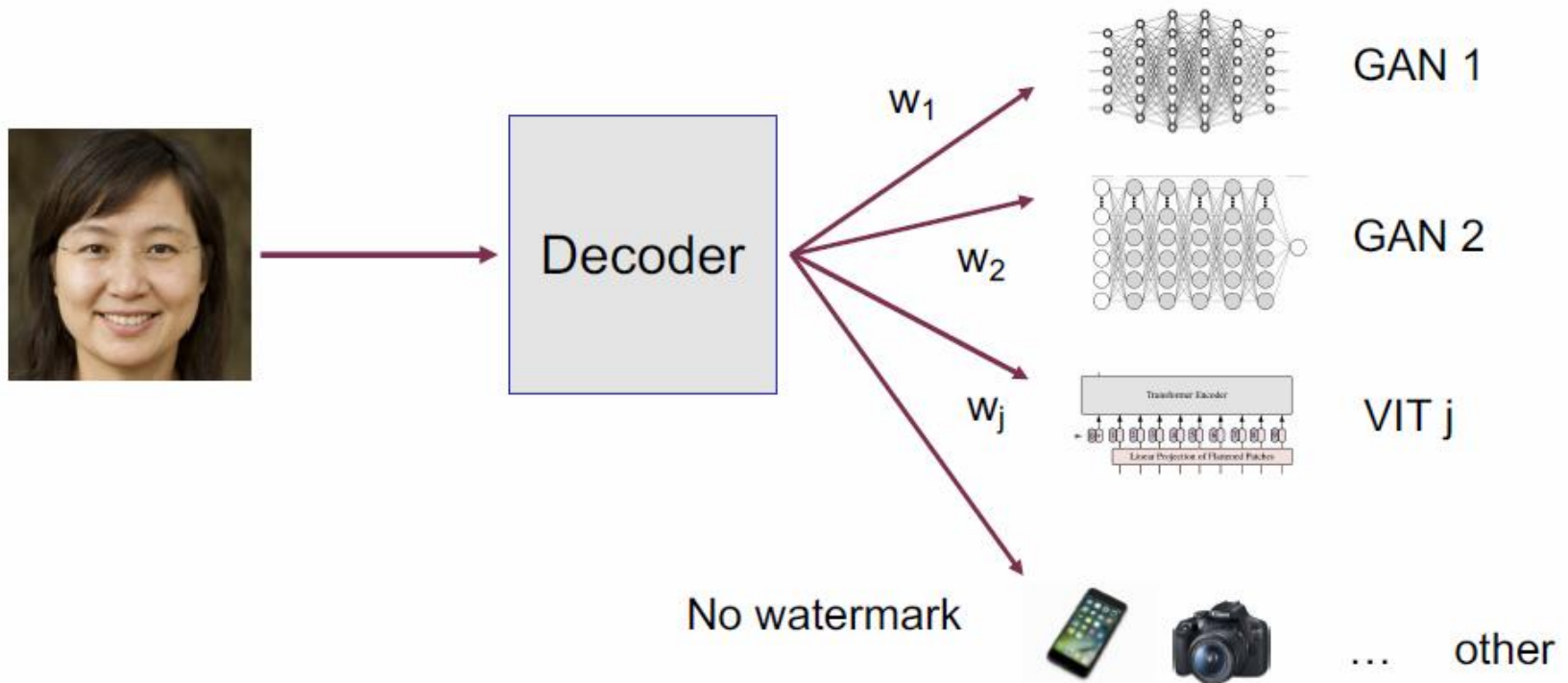
Box-free watermarking

Joint DNN and media watermarking

DNN watermarking

Kill two birds with one bullet

Box-free watermarking can be used to detect and trace synthetic contents to the model which generated them



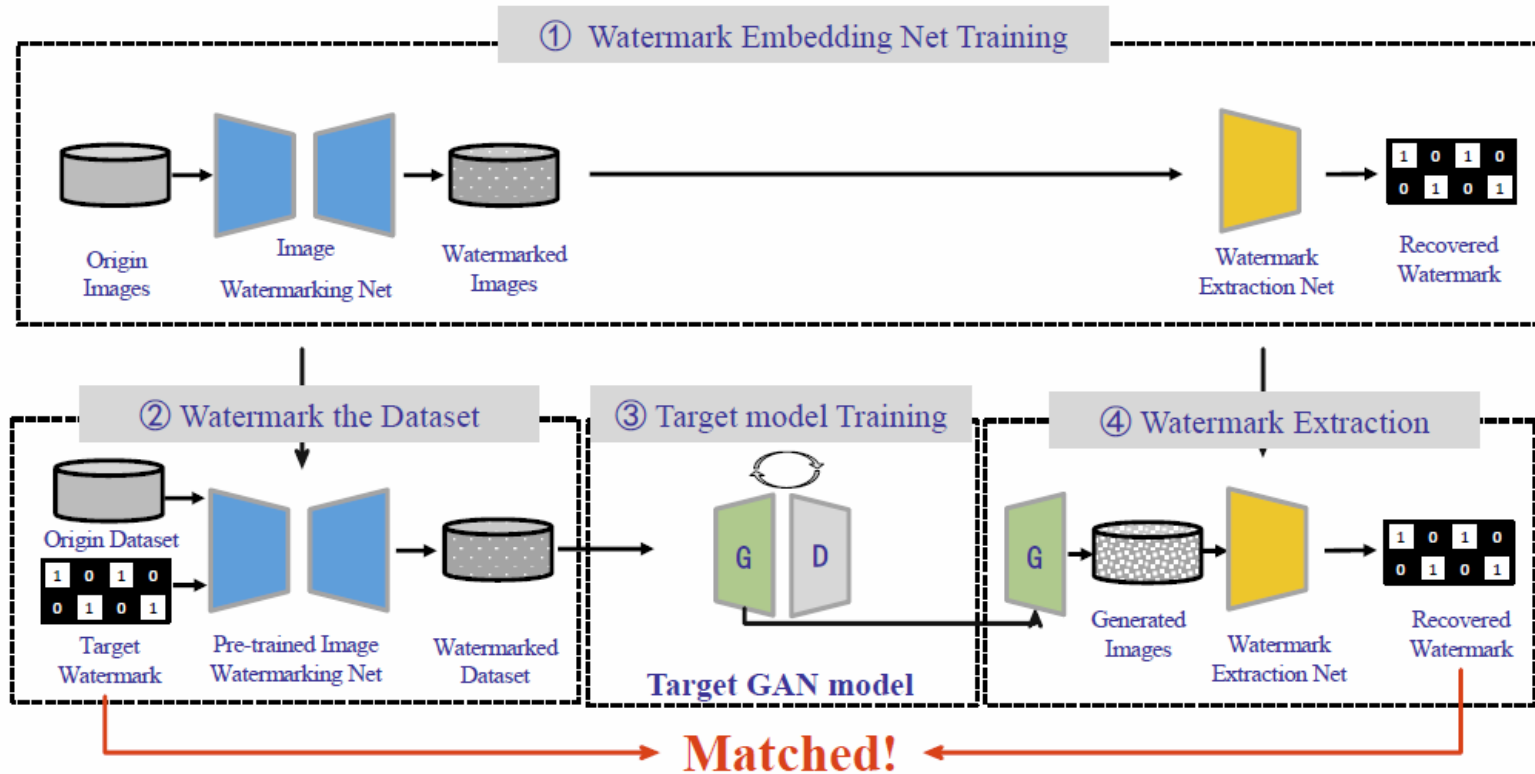
DNN watermarking

Challenges

- Development of a brand new theory of function watermarking
- Model-level robustness
- Image-level robustness
- Security model
 - Embedded information
 - Who does what?
 - Keyword management
- Security against intentional attacks

GAN watermarking: early and new solutions

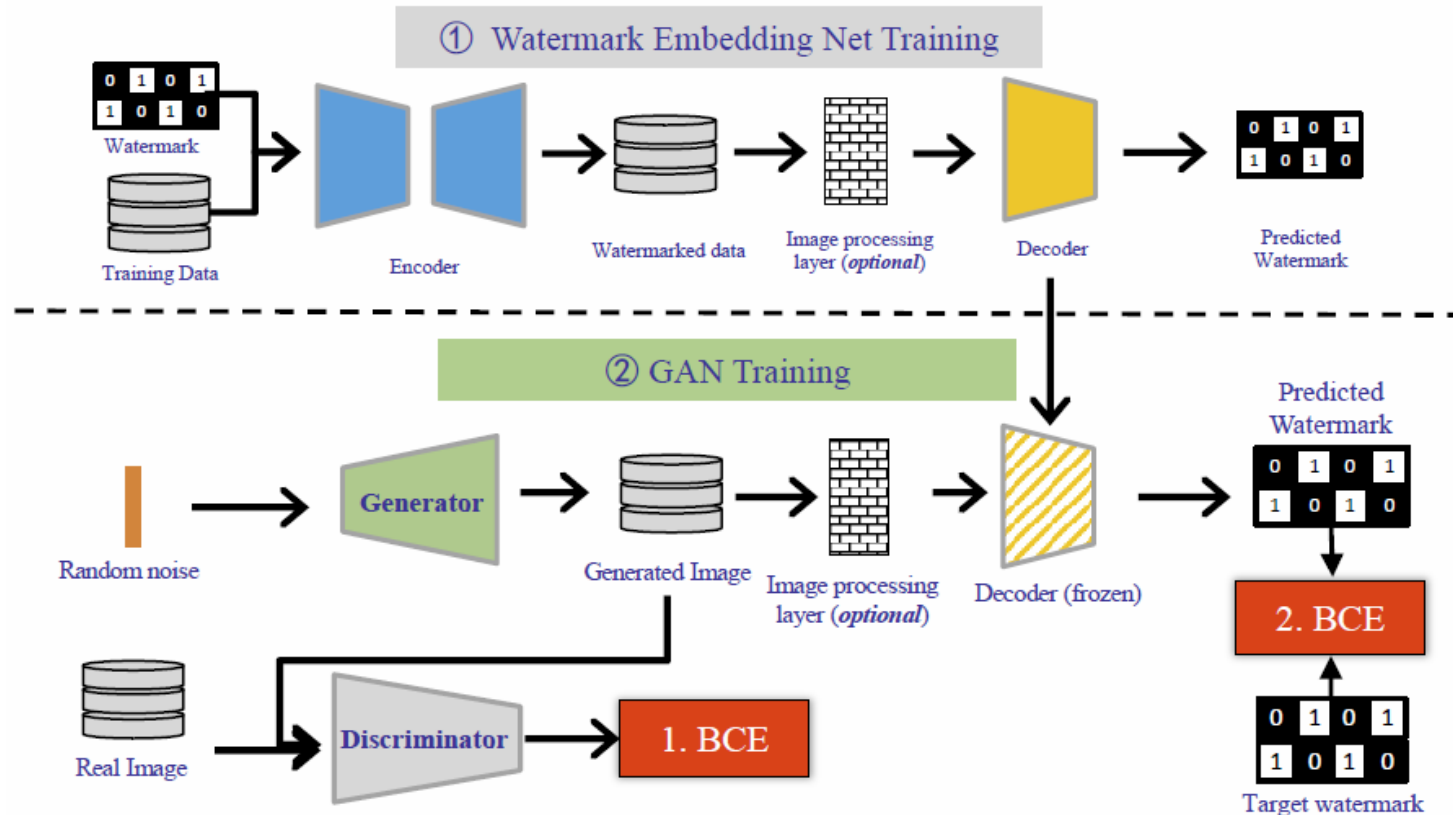
Early attempts: the imitation game



N. Yu, et. al, Artificial fingerprinting for generative models: Rooting deepfake attribution in training data, ICCV, 2021

GAN watermarking: early and new solutions

Supervised watermarking



Fei, J., Xia, Z., Tondi, B., & Barni, M. (2022, December). Supervised gan watermarking for intellectual property protection. In 2022 IEEE International Workshop on Information Forensics and Security (WIFS)

GAN watermarking: early and new solutions

Selected results

StyleGAN2 | FFHQ | 256

Clean
FID 5.28



Marked
FID: 6.16
Bits 100
Acc 99.75



Fei, J., Xia, Z., Tondi, B., & Barni, M. (2022, December). Supervised gan watermarking for intellectual property protection. In 2022 IEEE International Workshop on Information Forensics and Security (WIFS)

GAN watermarking: early and new solutions

Selected results: image processing



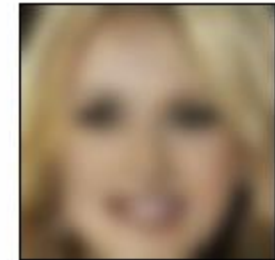
Original
Bit acc
100%



JPEG QF=30
Bit Acc 65%
PSNR 32.14



Noise Std=0.1
Bit Acc 77%
PSNR 20.05



Blurring k=3
Bit Acc 92%
PSNR 25.09



Brightness $\times 2.5$
Bit acc 78%
PSNR 14.97



Contrast $\times 2.5$
Bit acc 84%
PSNR 15.12



Saturation $\times 2.5$
Bit acc 98%
PSNR 14.96

GAN watermarking: early and new solutions

Selected results: model pruning



5%
Acc = **99.95%**



10%
Acc = **99.92%**



15%
Acc = **98.93%**



20%
Acc = **95.87%**



25%
Acc = **83.86 %**



30%
Acc = **67.90 %**



35%
Acc = **65.83 %**



40%
Acc = **61.15%**



50%
Acc = **57.09%**



60%
Acc = **52.35%**

GAN watermarking: early and new solutions

Selected results: model quantization



Precision 10^{-5}
Acc = **99.90%**



Precision 10^{-4}
Acc = **99.21%**



Precision 10^{-3}
Acc = **98.43%**



Precision 10^{-2}
Acc = **85.61%**



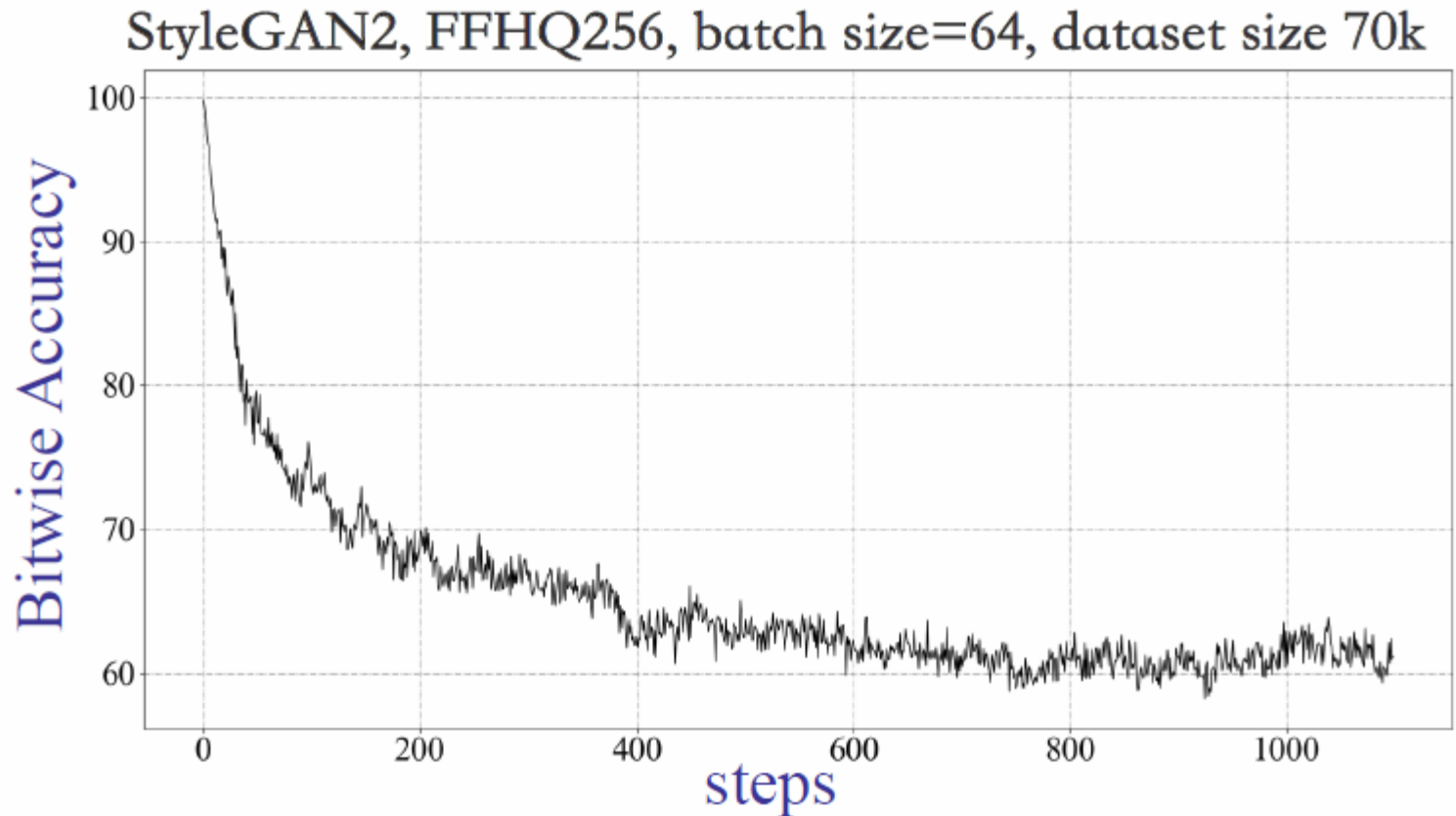
Precision 10^{-1}
Acc = **49.7%**



Precision 10^0
Acc = **53.52%**

GAN watermarking: early and new solutions

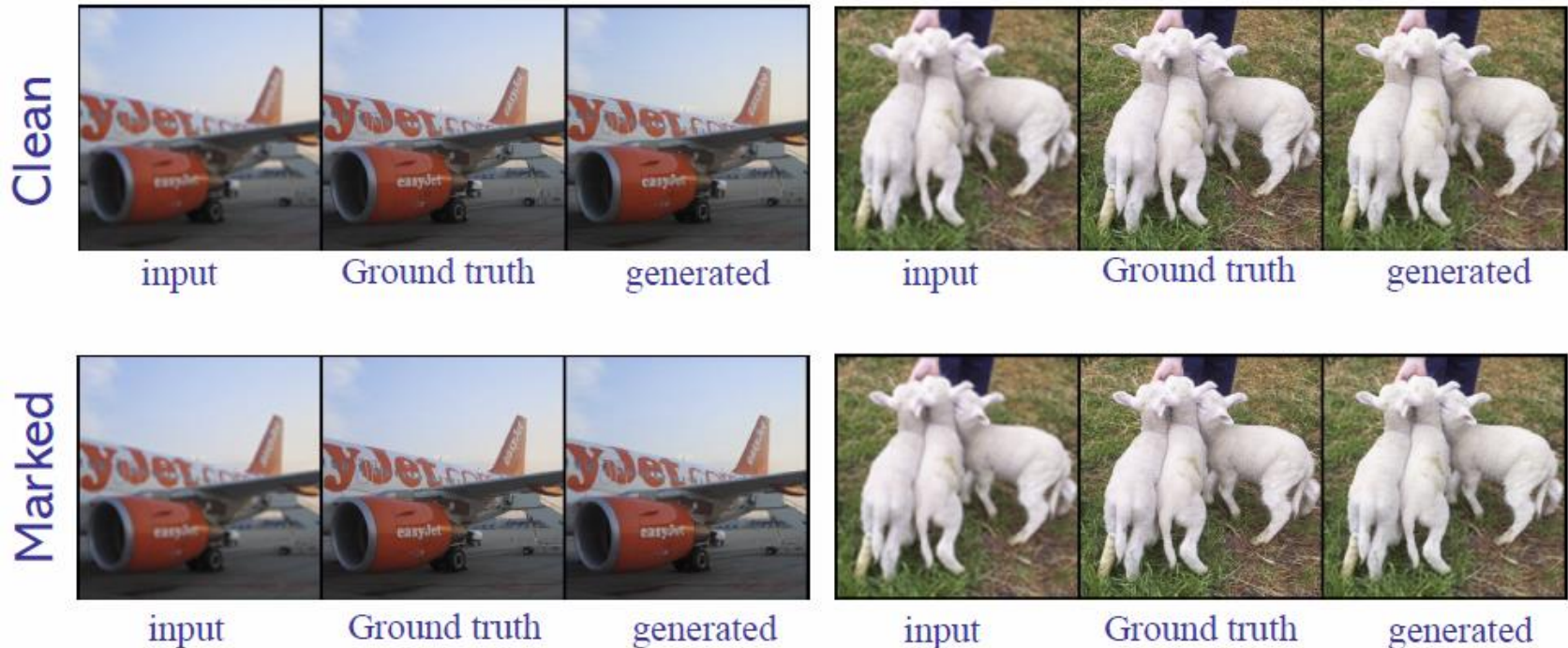
Selected results: fine tuning



Fine tuning attack without WM loss (1 epoch ~ 1093 steps)

GAN watermarking: early and new solutions

Selected results: super-resolution



PSNR 23.87 $\rightarrow$ 21.94 SSIM 0.7100 $\rightarrow$ 0.6206

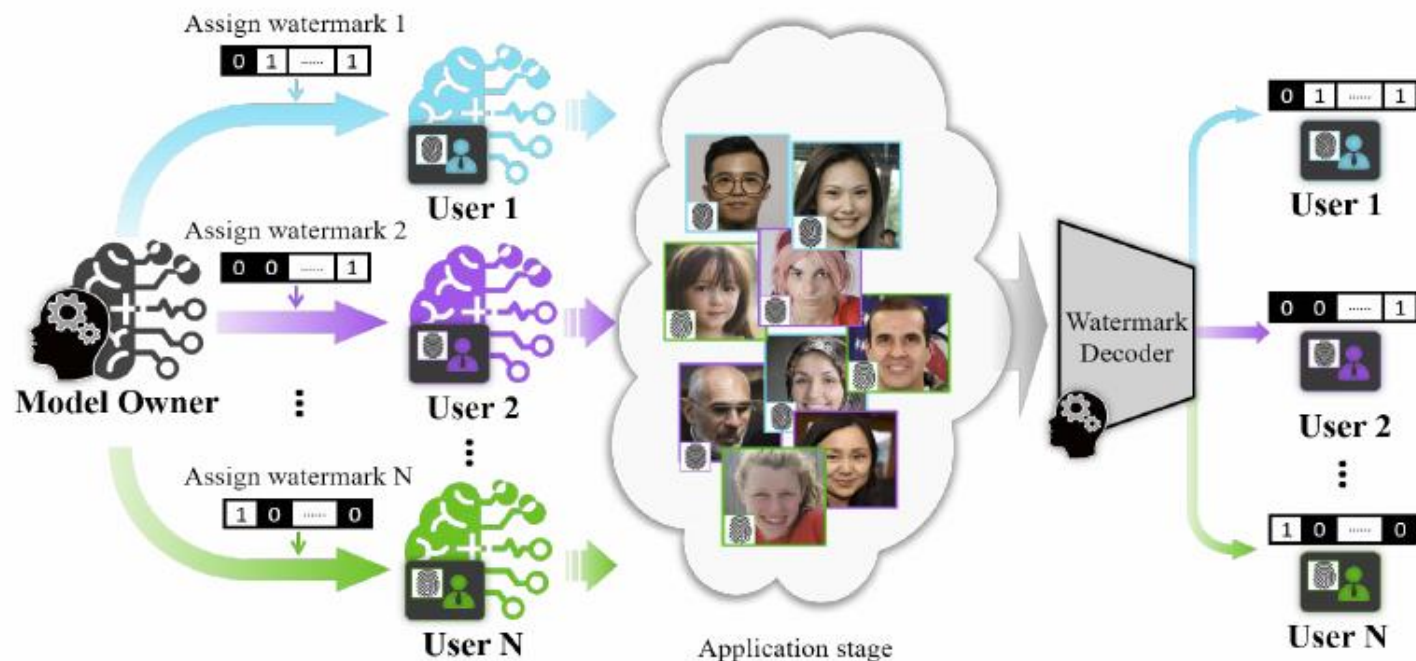
Fine tuning attack without WM loss (1 epoch $\sim$ 1093 steps)

GAN watermarking: early and new solutions

Retraining free fingerprinting

For some applications it may be useful to embed different watermarks in different version of the *same* DNN model

With most methods this requires a heavy retraining



GAN watermarking: early and new solutions

Solution

Introduce a personalized layer in the generator whose parameters are responsible for watermark embedding

The parameters are generated (feedforward) by a separate parameter-generation (*ParamGen*) network for each different watermark

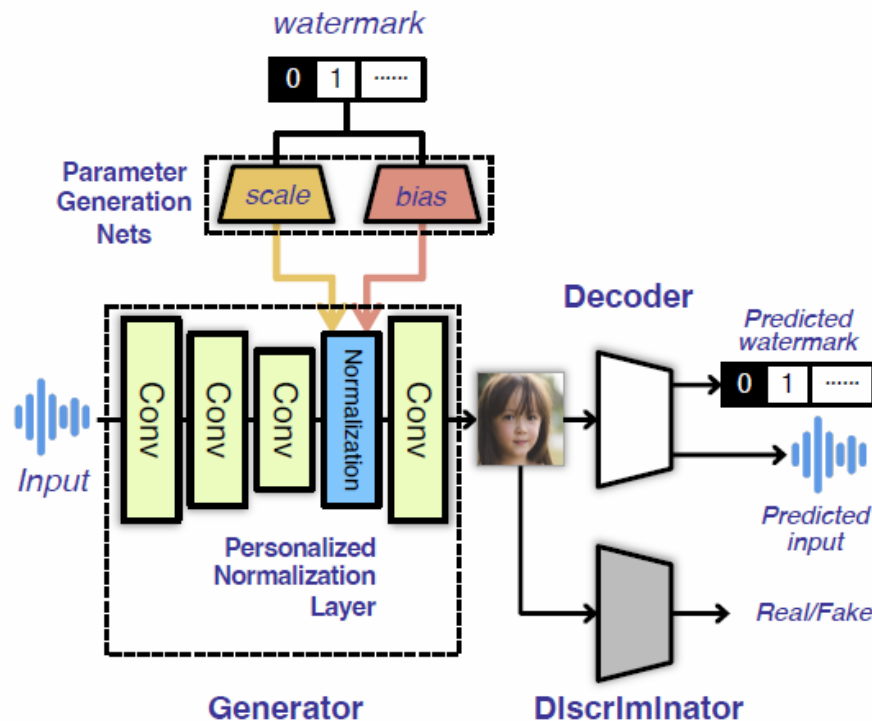


ParamGen network derives the parameters resulting in the embedding of the desired watermark without retraining

J. Zhang, D. Chen, J. Liao, W. Zhang, G. Hua, and N. Yu, “Passport-aware normalization for deep model protection,” Advances in Neural Information Processing Systems, vol. 33, pp. 22 619–22 628, 2020.

J. Fei, Z. Xia, B. Tondi, M. Barni, Robust retraining-free GAN fingerprinting via Personalized Normalization, IEEE WIFS 2023

GAN watermarking: early and new solutions



Losses

$$\mathcal{L}_{wat} = BCE(\text{watermark prediction})$$

$$\mathcal{L}_{adv} = BCE(\text{real/fake prediction})$$

$$\mathcal{L}_z = MSE(z, \text{predicted input})$$

$$\mathcal{L}_{const} = MSE(G_{w_1}(z), G_{w_2}(z))$$

Discriminator: $\mathcal{L}_{adv}$

Decoder: $\mathcal{L}_{wat}$ $\mathcal{L}_z$

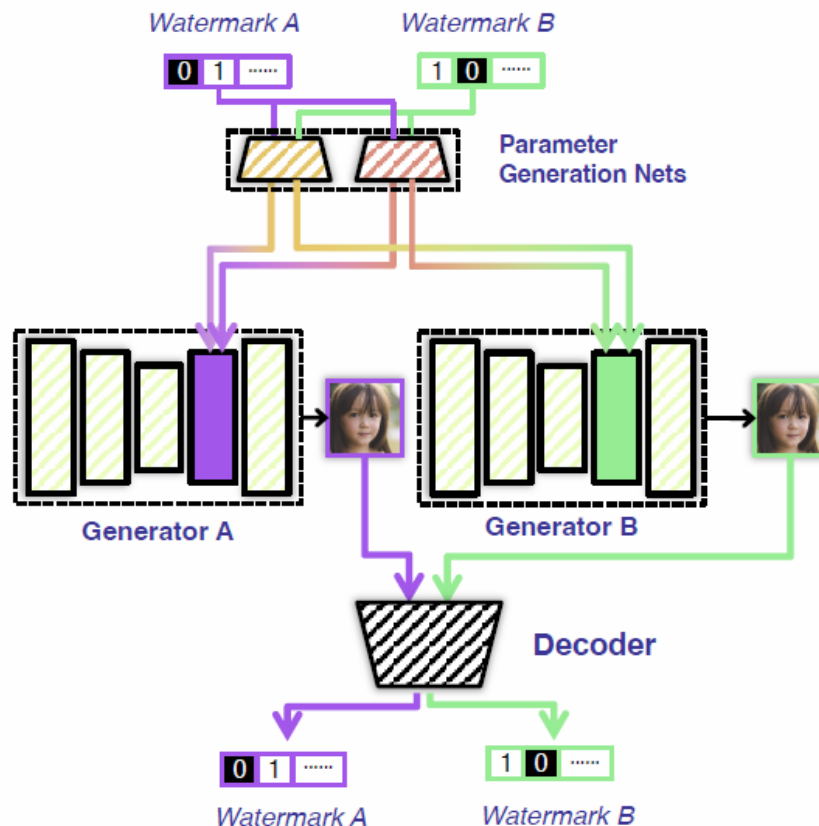
Generator, $\mathcal{L}_{adv}$ $\mathcal{L}_{const}$

ParamGen: $\mathcal{L}_z$ $\mathcal{L}_{wat}$

J. Fei, Z. Xia, B. Tondi, M. Barni, Robust retraining-free GAN fingerprinting via Personalized Normalization, IEEE WIFS 2023

GAN watermarking: early and new solutions

To create a model with a given watermark it is only necessary to run the ParamGen networks and generate the weights of the personalized normalization layer



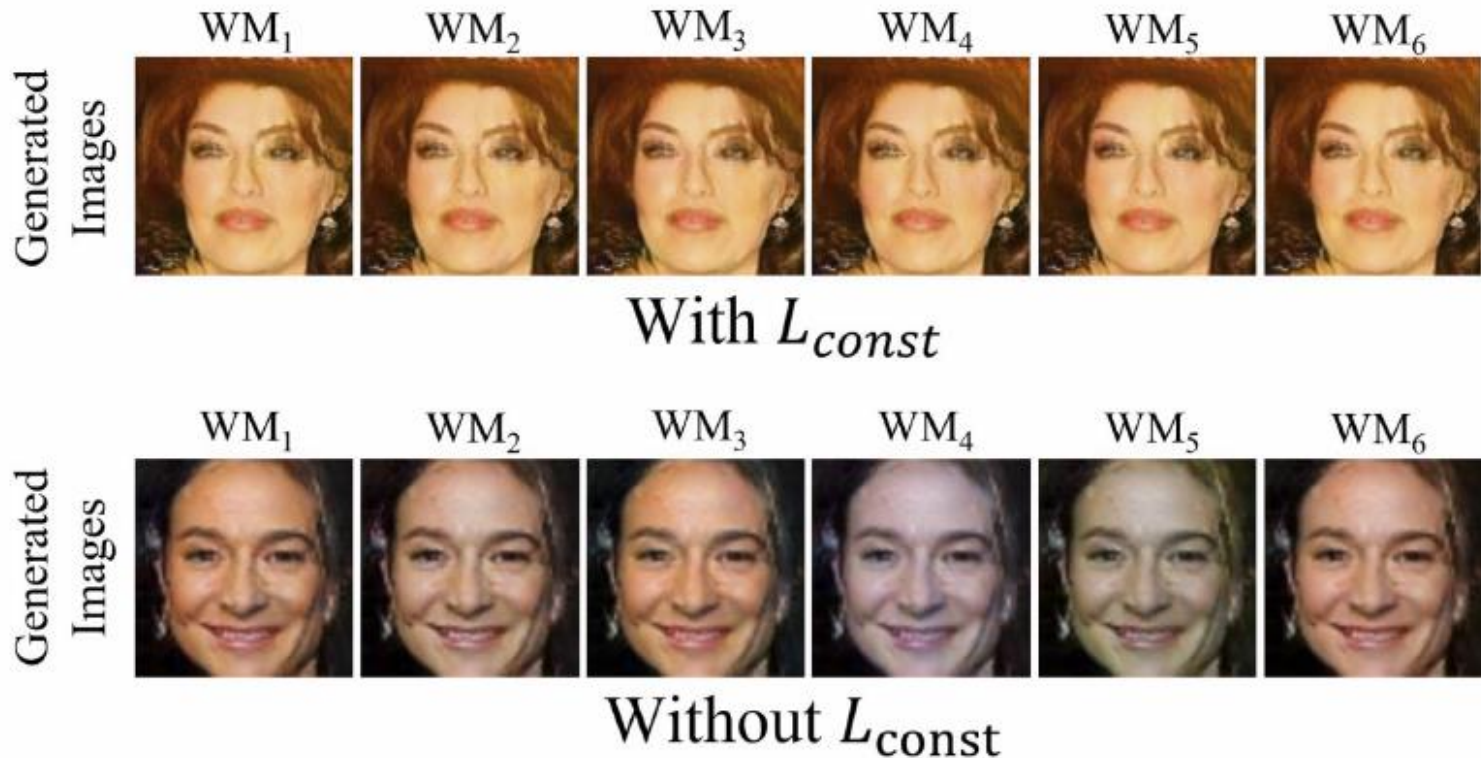
GAN watermarking: early and new solutions

Sense of achievable results

- Boundary Equilibrium GAN (BEGAN)
- Spectral Normalization GAN (SNGAN)
- Progressive Growing GAN (PGGAN)
- Face generation, trained on CelebA dataset
- Penultimate layer for PN
- ParamGen networks: fully connected, ReLu, 128 wm bits

GAN watermarking: early and new solutions

Impact of constant-loss term



The L_{const} is crucial to enforce a uniform behavior of models watermarked with different bits

GAN watermarking: early and new solutions

Achieved results

Model	Metric	No WM	Performance
BEGAN	Accuracy	-	100%
	FID	20.89	20.72
	Comput. overhead	-	100ms
SNGAN	Accuracy	-	99.9%
	FID	24.25	24.70
	Comput. overhead	-	100ms
PGGAN	Accuracy	-	99.8%
	FID	27.50	28.02
	Comput. overhead	-	100ms

GAN watermarking: early and new solutions

Achieved results

Robustness to model-manipulation

Model	Fine tuning	Pruning (10%)	Pruning (20%)	Quantization 10^{-1}
BEGAN	85%	99%	68%	98%
SNGAN	88%	98%	80%	99%
PGGAN	75%	99%	85%	100%

Robustness to image procesing

Model	JPEG (QF 50)	Blurring (5x5)	Noise addition (std = 0.1)
BEGAN	92%	80%	89%
SNGAN	92%	80%	90%
PGGAN	94%	84%	88%

Conclusions

- Limits of classical Multimedia forensics in the AI era
 - Still valid in specific, narrow, scenarios
 - Difficult (impossible) to cope with in a wide settings
 - Disinformation campaigns
- DNN-based active fingerprinting may provide a solution
 - Challenges to be solved (robustness and security)
 - No general solution, but can be a valid complement to passive MMF